| Plain Text | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency of alphabets in the plain text | 8 | 1 | 6 | 3 | 5 | 0 | 3 | 5 | 7 | 0 | 1 | 5 | 3 | 5 | 5 | 0 | 1 | 3 | 5 | 4 | 3 | 0 | 4 | 0 | 2 | 0 |

With the help of Cipher table and frequency table, let us fill the Cipher numbers for the plain text. Let us first match the most repeated letters (5 times and above), then 2 to 4 times repeated letters and finally remaining letters step by step as shown in given figure.

**Step 1:- Cipher number for most repeated letters(5 times and above)**

| Plain text | m | a | t | h | e | m | a | t | i | c | s | | i | s | | a | | u | n | i | q | u | e | | s | y | m | b | o | l | i | c |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | | 05 | | 12 | 09 | | 05 | | 13 | 07 | 23 | | 13 | 23 | | 05 | | | 18 | 13 | | | 09 | | 23 | | | | 19 | 16 | 13 | 07 |

| Plain text | l | a | n | g | u | a | g | e | | i | n | | w | h | i | c | h | | t | h | e | | w | h | o | l | e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | 16 | 05 | 18 | | | 05 | | 09 | | 13 | 18 | | | 12 | 13 | 07 | 12 | | | 12 | 09 | | | 12 | 19 | 16 | 09 |

| Plain text | w | o | r | l | d | | w | o | r | k | s | | a | n | d | | a | c | t | s | | a | c | c | o | r | d | i | n | g | l | y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | | 19 | | 16 | | | | 19 | | | 23 | | 05 | 18 | | | 05 | 07 | | 23 | | 05 | 07 | 07 | 19 | | | 13 | 18 | | 16 | |

**Step 2: Cipher number for most repeated letters (2 times and above)**

| Plain text | m | a | t | h | e | m | a | t | i | c | s | | i | s | | a | | u | n | i | q | u | e | | s | y | m | b | o | l | i | c |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | 17 | 05 | 24 | 12 | 09 | 17 | 05 | 24 | 13 | 07 | 23 | | 13 | 23 | | 05 | | 25 | 18 | 13 | | 25 | 09 | | 23 | 03 | 17 | | 19 | 16 | 13 | 07 |

| Plain text | l | a | n | g | u | a | g | e | | i | n | | w | h | i | c | h | | t | h | e | | w | h | o | l | e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | 16 | 05 | 18 | 11 | 24 | 05 | 11 | 09 | | 13 | 18 | | 01 | 12 | 13 | 07 | 12 | | 24 | 12 | 09 | | 01 | 12 | 19 | 16 | 09 |

| Plain text | w | o | r | l | d | | w | o | r | k | s | | a | n | d | | a | c | t | s | | a | c | c | o | r | d | i | n | g | l | y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | 01 | 19 | 22 | 16 | 08 | | 01 | 19 | 22 | | 23 | | 05 | 18 | 08 | | 05 | 07 | 24 | 23 | | 05 | 07 | 07 | 19 | 22 | 08 | 13 | 18 | 11 | 16 | 03 |

**Step 3: Cipher numbers for non-repeated letters**

| Plain text | m | a | t | h | e | m | a | t | i | c | s | | i | s | | a | | u | n | i | q | u | e | | s | y | m | b | o | l | i | c |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | 17 | 05 | 24 | 12 | 09 | 17 | 05 | 24 | 13 | 07 | 23 | | 13 | 23 | | 05 | | 25 | 18 | 13 | 21 | 25 | 09 | | 23 | 03 | 17 | 06 | 19 | 16 | 13 | 07 |

| Plain text | l | a | n | g | u | a | g | e | | i | n | | w | h | i | c | h | | t | h | e | | w | h | o | l | e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | 16 | 05 | 18 | 11 | 25 | 05 | 11 | 09 | | 13 | 18 | | 01 | 12 | 13 | 07 | 12 | | 24 | 12 | 09 | | 01 | 12 | 19 | 16 | 09 |

| Plain text | w | o | r | l | d | | w | o | r | k | s | | a | n | d | | a | c | t | s | | a | c | c | o | r | d | i | n | g | l | y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cipher Numbers | 01 | 19 | 22 | 16 | 08 | | 01 | 19 | 22 | 15 | 23 | | 05 | 18 | 08 | | 05 | 07 | 24 | 23 | | 05 | 07 | 07 | 19 | 22 | 08 | 13 | 18 | 11 | 16 | 03 |

Thus, the Additive Cipher text for plain text is as follows:

**"17 05 24 12 09 17 05 24 13 07 23 13 23 05 25 18 13 21 25 09 23 03 17 06 19 16 13 07 16 05 18 11 25 05 11 09 13 08 01 12 13 07 12 24 12 09 01 12 19 16 09 01 19 22 16 08 01 19 22 15 23 05 18 08 05 07 24 23 05 07 07 19 22 08 13 18 11 16 03"**



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